EXPERIMENT 25:

Demonstrate Infrastructure as a Service (IaaS) by creating a resources group using a Public Cloud Service Provider (Azure), configure with minimum CPU, RAM, and Storage.

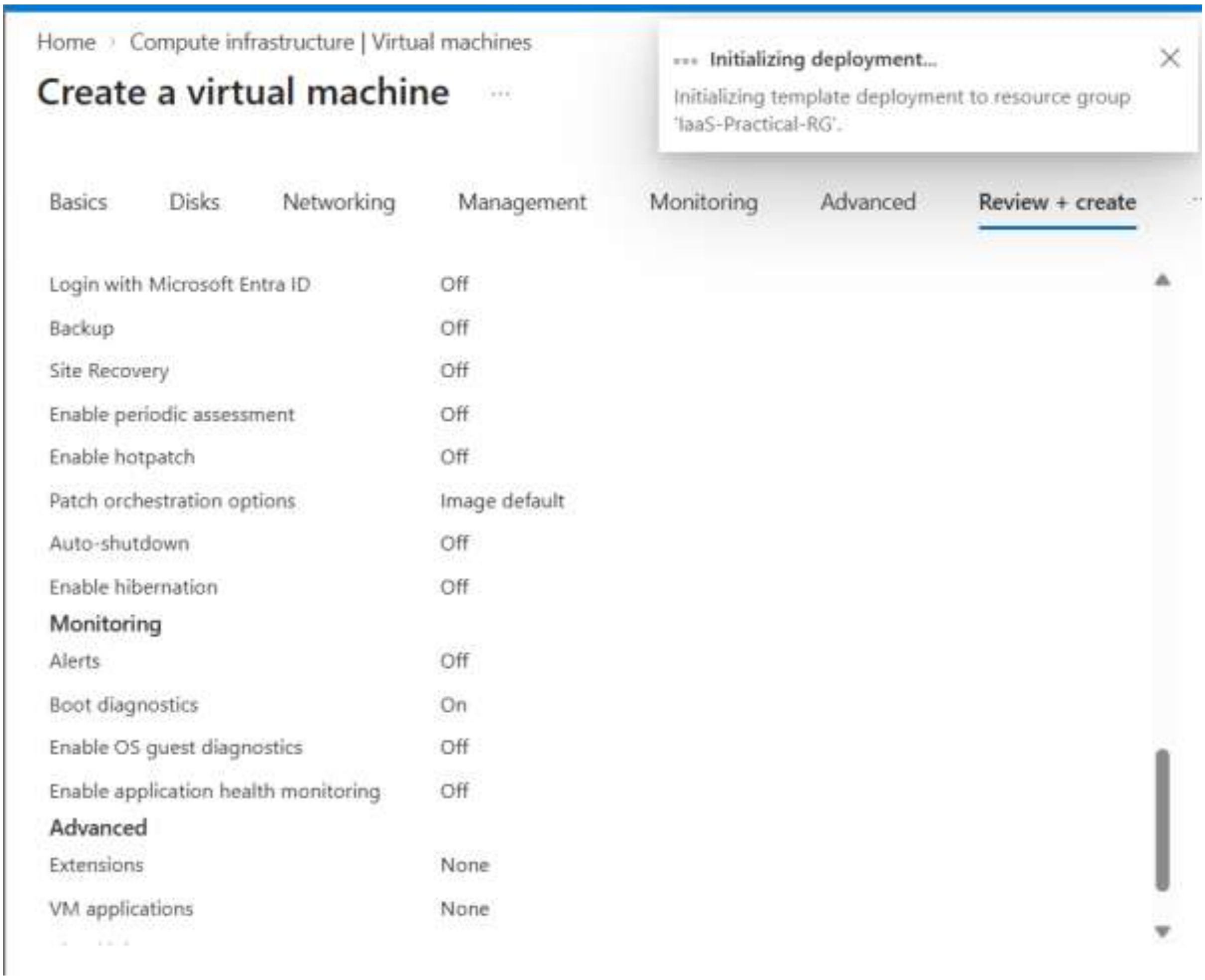
Aim

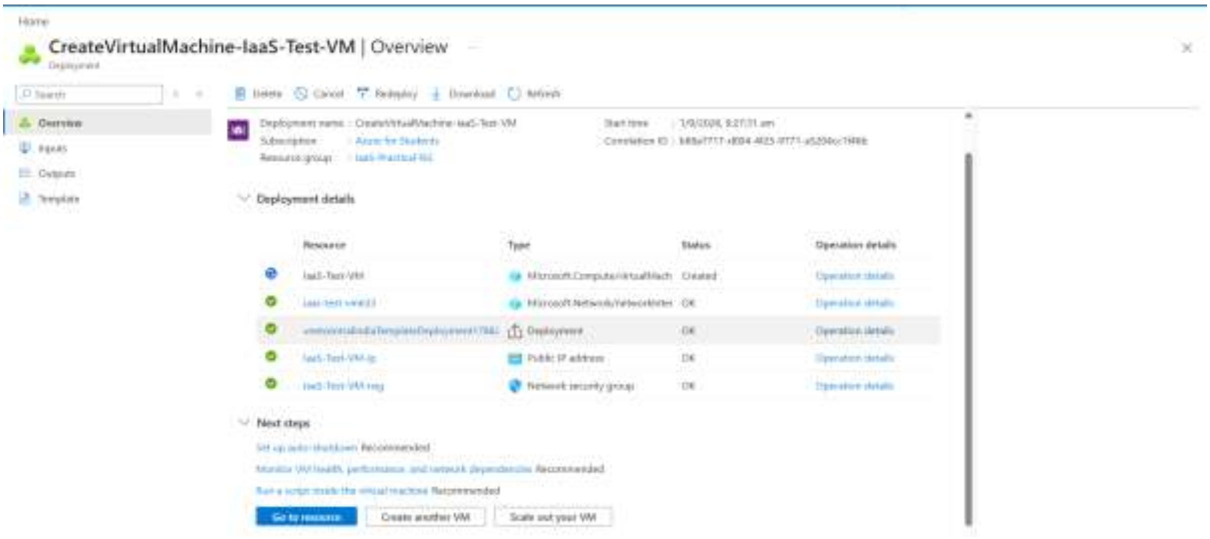
To demonstrate Infrastructure as a Service (IaaS) using Microsoft Azure by creating a Resource Group and deploying a lightweight virtual machine with minimum available CPU, RAM, storage, and networking resources.

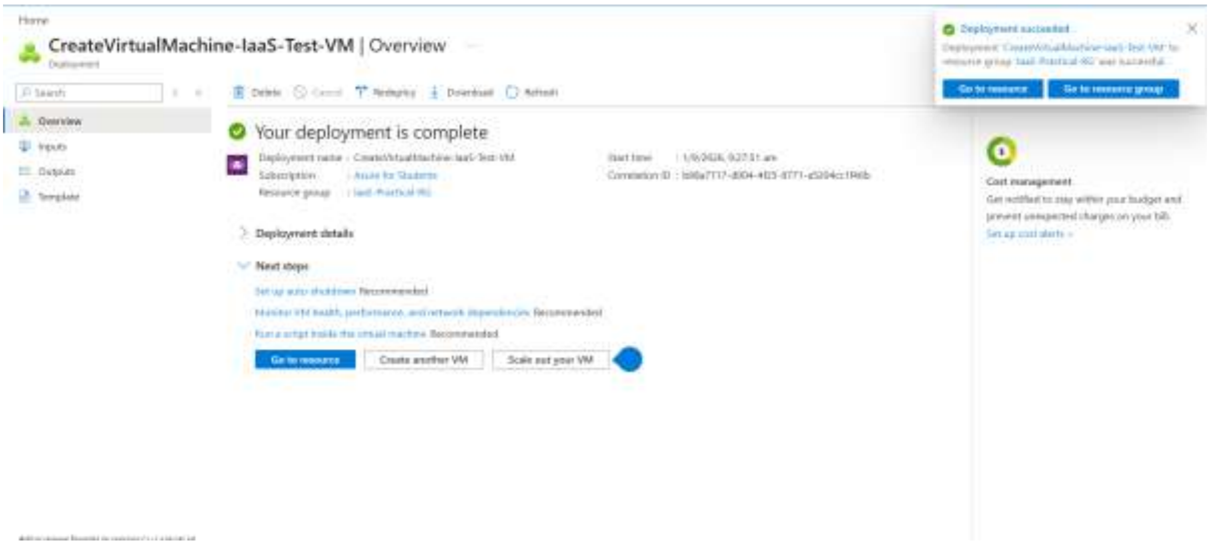
Procedure

1. Open the Microsoft Azure Portal and sign in using the Azure account.
2. Search for Resource Groups, select Create, choose the required subscription and region, and create a Resource Group named IaaS-Practical-RG.
3. Search for Virtual Machines, select Create → Azure Virtual Machine, and select the previously created Resource Group.
4. Enter the virtual machine name as IaaS-Test-VM and select Ubuntu Linux as the operating system image.
5. In the Size section, select the smallest suitable VM available. From the available options, select B2ats_v2, which provides 2 vCPUs and 1 GiB RAM.
6. In the Disks section, select the minimum suitable OS disk and do not add additional data disks. Configure the required virtual network, subnet, public IP, and network security group under Networking.
7. Configure an administrator username and SSH authentication, review all settings, and click Create to deploy the virtual machine.
8. After deployment, open the VM's Overview page and verify the Resource Group, VM size, operating system, CPU, RAM, storage, and running status.
9. Connect to the Ubuntu VM using SSH and verify the resources using commands such as lscpu, free -h, and df -h.
10. Take screenshots of the Resource Group, VM configuration, selected B2ats_v2 size, and resource verification commands as evidence of the implementation.

IMPLEMENTATION:







Result

The IaaS environment was successfully created using Microsoft Azure. A Resource Group and Ubuntu Virtual Machine were deployed with the minimum available configuration of 2 vCPUs, 1 GiB RAM, and minimum suitable storage. The virtual machine and its allocated resources were successfully verified through the Azure Portal and Linux system commands.